

Capacitated single-allocation hub location model for a flood relief distribution network

Ornurai Sangsawang^{ID} | Sunarin Chanta^{ID}

Department of Industrial Management,
King Mongkut's University of Technology
North Bangkok, Prachinburi, Thailand

Correspondence

Ornurai Sangsawang, Department of
Industrial Management, King Mongkut's
University of Technology North Bangkok,
Prachinburi, Thailand.
Email: ornurai.s@fitm.kmutnb.ac.th

Abstract

In disaster management, the logistics for disaster relief must deal with uncontrolled variables, including transportation difficulties, limited resources, and demand variations. In this work, an optimization model based on the capacitated single-allocation hub location problem is proposed to determine an optimal location of flood relief facilities with the advantage of economies of scale to transport commodities during a disaster. The objective is to minimize the total transportation cost, which depends on the flood severity. The travel time is bounded to ensure that survival packages will be delivered to victims in a reasonable time. Owing to complexity of the problem, a hybrid algorithm is developed based on a variable neighborhood search and tabu search (VNS-TS). The computational results show that the VNS found the optimal solutions within a 2% gap, while the proposed VNS-TS found the optimal solution with a 0% gap. A case study of severe flooding in Thailand is presented with consideration of related parameters such as water level, hub capacity, and discount factors. Sensitivity analyses on the number of flows, discount factors, capacity, and bound length are provided. The results indicated that demand variation has an impact on the transportation cost, number of hubs, and route patterns.

KEYWORDS

flood, hub location, optimization, Tabu search, variable neighborhood search

1 | INTRODUCTION

Natural disasters lead to human suffering, especially in situations in which many victims need assistance at the same time under uncontrollable emergency conditions. Examples of such disasters include earthquakes, floods, droughts, hurricanes, and epidemics. To relieve the suffering, assistance such as survival packages, medications, and rescue teams need to be distributed as rapidly as possible. In this case, the transportation structure is a key factor affecting the delivery time. Generally, assistance is provided by many organizations, for example, public and private organizations and government agencies, while the needs of victims are also spread out over the affected areas. Thus, independent shipments from an origin to a destination can lead to high costs and unusually long delivery times. To speed up the delivery process, we can develop a design to establish hubs where supplies from different sources can be grouped and distributed in large quantities for delivery to the final destinations. These hubs serve as relief centers and have functions including receiving supplies, determining personnel needs, sorting packages, and distributing packages to victims. Many significant factors affect relief operations, such as the number of volunteers, number of victims, available equipment, and locations of victims in affected areas, including the locations of relief centers. During disasters in particular, it is necessary to manage these uncontrolled factors against the limitation of available resources and difficulty of transportation. Unlike in a business network flow, in which the overall system cost is a priority and those in remote areas are left behind, a focus on the sufferers' satisfaction and travel assurance capability should be imposed in disaster management.¹

In this study, we determine the locations of relief centers using a hub location problem (HLP), in which nodes represent sources of help and destinations of needs, and hubs represent relief centers. A transportation network during a flood disaster is created by establishing hubs and assigning nodes to the hubs to minimize total transportation costs. The HLP is a type of facility location problem that aims to identify the locations of various facilities such as warehouses,² service centers,³ and rescue units.⁴ There are several different types of hub problems. The differences are based on related factors such as the capacity of the facilities, number of hubs to be established, number of hubs allowed to connect with nodes, and model objective. The capacitated single-allocation HLP (CSAHL) limits the capacity of the facilities, which is the opposite of the uncapacitated single-allocation hub location problem (USAHL), which does not consider the facility capacities. In the uncapacitated single-allocation p -hub location problem (USApHLP) and the capacitated single allocation p -hub location problem (CSApHLP), the number of hubs is already known or fixed as equal to p . The demand node assignment to hub locations is commonly classified into two types: the uncapacitated multiple-allocation p -hub location problem (UMApHLP) and the capacitated multiple-allocation p -hub location problem (CMApHLP). Demand nodes may route through either a single hub or several hubs. Moreover, there are specific types of hub location problems based on the objective function employed, such as the uncapacitated single allocation p -hub median problem (UMApHMP) and the uncapacitated single allocation p -hub center problem (UMApHCP). Different types of hub location problem models are detailed in Farahani et al⁵ and Mahmutogullari and Kara.⁶ However, these general models are applicable for typical transportation problems only and not for transportation during a flood disaster. Thus, we developed an appropriate model for this particular situation, which considers the limitations and involved variable changes.



We propose an extended version of the CSAHLP, in which the number of hubs is unknown, the demand nodes are routed to single hubs, and the hubs have a limited capacity. The optimal number of hubs is investigated by considering the fixed costs of locating the hub. The optimal routes from sources to destinations are assigned by considering the travel cost between pairs. Important factors that affect transportation during flood disasters are also considered, such as different water levels, the maximum distance from nodes to hubs, and limitation of hub capacities. The number of variables and large scales of real problems can further complicate the issue. In addition, hub location problems are *NP*-hard problems, consisting of two subproblems: locating the hubs and allocating nodes to hubs,⁷ which can be solved in polynomial time.

In this case, an exact method is not practical. Thus, we develop a hybrid algorithm, the variable neighborhood search with tabu search (VNS-TS), to solve the problem. The performance of the algorithm is validated by a comparison with the VNS using Australian Post (AP) data. The proposed algorithm is then applied to solve a real-world flooding case study in Thailand. The proposed algorithm shows an efficient performance with acceptable solving time and high solution quality.

The remainder of this article is organized as follows. A brief review of related literature is provided in Section 2, followed by the details and formulations of the proposed model in Section 3. Next, we describe the details of the VNS and its integration with the TS in Section 4. In Section 5, experiments are conducted to test the performance of the proposed algorithm using the standard problem, and then in Section 6, we present a real-world case study using data collected from the severe flooding in Thailand in 2011. In Section 7, a sensitivity analysis of the related factors is provided. Finally, conclusions are discussed with a direction for future research.

2 | LITERATURE REVIEW

2.1 | Applications of hub location problems

Hub location problems are applied to solve distribution problems in many systems such as airlines,⁸⁻¹¹ postal networks,¹²⁻¹⁴ and transportation systems.¹⁵⁻¹⁸

The basic structure of hub location problems and network designs is composed of nodes and hubs, where commodities are shipped from node to node via hubs. We take advantage of the economies of scale by transferring flows between hubs. In transportation problems, there are special types of hub locations for which the formulations are adjusted to solve different situations. Berman et al¹⁹ investigated the location of a facility and several transfer points, in which the transfer points serve as collection points of demands that needed the service from a central facility. Three heuristics, the decent approach, simulated annealing (SA), and TS, were developed to solve the problem. The computational results showed that SA performed the best among these three heuristics. Chong et al²⁰ considered a problem of scheduling and routing shipments in a hybrid hub network, in which direct delivery was allowed, and a set of intershipment times was given. The objective was to minimize the transportation and inventory holding costs. The multiple frequency hybrid system (MFHS) heuristic procedure was proposed to solve the problem. The procedure starts by decomposing the network into a series of single links with a single supplier and customer and then searches for possible improvements in the network to find the best route. Correia et al²¹ studied a capacitated single-assignment hub location problem with multiple



capacity levels, in which the size of the hubs was a decision variable. They proposed formulations and valid inequalities to solve the problem. Kartal et al²² proposed a single allocation p -hub median location and routing problem, for which they considered the routing between nodes in a network. Two heuristics were proposed: the multi-start SA and ant colony system (ACS).

One important factor that has to be considered in transportation is response time, especially during a disaster. Because the objective is to provide timely help, some studies focus on the response time or time to transfer supplies from hub to node in addition to the design of the relief network. Yaman and Elloumi²³ proposed a star p -hub center problem and a star p -hub median problem with bounded path lengths for designing a telecommunication network. Hasan-zadeh and Bashiri²⁴ designed a relief logistics network for disaster preparedness by considering emergency conditions in which relief operations can be performed for a node if it is inside of the relief radius. The proposed model is based on the hub covering location problem, and Lagrangian relaxation was selected to solve the problem. Xing²⁵ presented an emergency material collecting model of a disaster with a fuzzy collecting time, capacity limitation, and time constraint on transferring, and a particle swarm optimization (PSO) algorithm was developed to solve the problem. Martins de Sá et al²⁶ improved the service level of a multiple allocation incomplete hub location problem by adding a service time requirement. The problem was solved by using the Benders decomposition algorithm. Khodemani-Yazdi et al²⁷ presented a bi-objective hierarchical hub location problem to locate servicing centers, with the objective of minimizing the total cost and maximum route length. A hybrid SA was proposed to solve the problem.

In most studies of hub location problems, the parameters are assumed to be certain values under usual conditions. For example, travel time is directly proportional to distance; however, in practice, it might vary under different conditions such as environment, traffic, and road conditions. Many works have been proposed to estimate travel time and other parameters considering the operational environment. Yang et al²⁸ considered the p -hub center problem with an uncertain travel time characterized by normal fuzzy vectors for the delivery of perishable or time-sensitive systems. The objective was to minimize the maximum travel time between any origin-destination pair. A hybrid algorithm, incorporating a genetic algorithm (GA) and local search (LS), called GALS, was developed to solve the problem, and it was tested on simulated problems. The results showed that GALS performed better than the standard GA in terms of solution and runtime. Sadeghi et al²⁹ proposed a stochastic approach for a reliable p -hub covering location problem, in which the objectives were to minimize the total transportation cost and to obtain the maximum flows carried by the network, when its link capacities depended on stochastic conditions, such as daily traffic and disaster. The differential evolution (DE) algorithm was developed to solve the problem. Gao and Qin³⁰ modeled the p -hub center problem and considered travel time as an uncertain variable based on expert subjective belief in the case of a lack of data. Alumur et al³¹ considered service time and congestion at the hub. The service time refers to the travel time between the node and hub and serving time at the hub, where the serving time depends on the congestion level at the hub. Mohammadi et al³² investigated single-allocation p -hub location problems and the effect of uncertainties on deliveries. A hybrid algorithm, composed of a GA and VNS, was proposed to solve the problem.

In our problem, we consider parameters affected by floodwater levels such as transportation cost, fixed cost, and travel time between nodes and hubs. In the case of disaster situations especially, victims may not receive relief packages in time, which has an effect on their lives. These factors are considered in the proposed model.



2.2 | Solution methods for the hub location problem

The hub location problem is known as an *NP*-hard problem, which means it is difficult to solve or can be solved in polynomial time. Therefore, these types of problems require efficient solution methods. Here, we address related works of single hub location problems by highlighting the methodology that has been developed for solving them. One well-recognized hub location problem is the uncapacitated single allocation *p*-hub median problem (USApHMP), in which the objective is to minimize the total transportation cost, while the costs of establishing hubs are ignored. The problem was first proposed as a quadratic integer formulation by O'Kelly.³³ Several heuristic approaches, such as a TS by Klincewincz³⁴ and Skorin-Kapov and Skorin-Kapov³⁵; SA by Ernst and Krishnamoorthy³⁶; path-relinking (PR) by Pérez et al³⁷; GA by Kratica et al³⁸; and general VNS by Ilić et al,³⁹ have been developed to solve the USApHMP. Recently, for solving the CSAHLP, Marić et al⁴⁰ proposed a memetic algorithm (MA), in which two efficient local searches were used to improve both the location and allocation parts of the problem. In addition, Abyazi-Sani and Ghanbari⁴¹ proposed an efficient TS, in which the TS computes the changes in the objective function's value in each move.

The single allocation hub location problem with fixed costs was formulated by O'Kelly.⁴² Campbell⁴³ developed a linear integer formulation for solving the capacitated location problem. Ernst and Krishnamoorthy⁴⁴ presented a mixed integer linear programming method for solving capacitated single allocation hub location problems. The SA and random descent heuristic were also proposed. For small-sized problems, the random descent heuristic was comparable. However, the SA was comparable for large-sized problems. Ernst's algorithms can solve problems with sizes up to 50 nodes. Yaman and Carello⁴⁵ considered the cost of using an edge as a stepwise function and had a capacity constraint on both the incoming and outgoing flows at hubs. They proposed a heuristic method that divided the problem into subproblems: location and assignment, in which the location subproblem was solved by TS, and then the assignment subproblem was solved by a greedy algorithm. Stanimirović⁴⁶ and Kratica et al⁴⁷ proposed a GA for solving the CSAHLP, in which a caching technique was applied to improve the computational time. Their GAs showed a good performance on both small and large problems. Randall⁴⁸ compared four types of the ant colony optimization (ACO) with SA for solving the CSAHLP, and their results showed that ACOs performed significantly better than SA in terms of the objective values for small-sized problems. Hoff et al⁴⁹ considered the CSAHLP with modular link capacities, in which the capacity of the edges between hubs was increased in a modular way. They applied a memory structure to the well-known TS, in which PR was applied as a post-process, and the experiments showed that the memory structure performed well. Yang et al⁵⁰ developed an MA to solve the CSAHLP with routing problem, in which collection and delivery tours were considered.

There are significant differences between previous studies and our work. Table 1 provides a summary of previous works, including the model objective, constraint consideration, and solution method. Note that we listed related HLP models, which are the USApHMP and CSAHLP. The proposed model considers an important issue of flooding, the height of the water level. This has an effect on other parameters such as travel time, cost, and type of vehicles; therefore, the model can capture the real characteristics of the logistics during a flood disaster. As our goal is to solve real-world problems, we have to consider a large-scale problem and a large number of variables, which make the problem more complex and difficult to solve. Therefore, we need to construct an efficient solution method. Based on the previous works reviewed, we developed a hybrid algorithm, VNS-TS, which combined two solution methods that showed high performance

TABLE 1 Summary of previous works on hub location problems (USApHMP and CSAHLP)

Authors	Year	Objective		Consideration				Solution method
		Min travel cost	Min establish cost	Hub capacity	Path length	Travel time variation	Height of flood	
Klincewincz ³⁴	1992	✓						TS
Skorin-Kapov et al ³⁵	1994	✓						TS
Abyazi-Sani and Ghanbari ⁴¹	2016	✓						TS
Ernst and Krishnamoorthy ³⁶	1996	✓						SA
Pérezet al ³⁷	2007	✓						PR
Kratika et al ³⁸	2007	✓						GA
Ilić et al ³⁹	2010	✓						VNS
Marić et al ⁴⁰	2013	✓						MA
Ernst and Krishnamoorthy ⁴⁴	1999	✓	✓	✓				SA
Yaman and Carello ⁴⁵	2005	✓	✓	✓				TS, Greedy Algorithm
Stanimirović ⁴⁶	2007	✓	✓	✓				GA with caching technique
Kratika et al ⁴⁷	2011	✓	✓	✓				GA with caching technique
Randall ⁴⁸	2008	✓	✓	✓				ACO, SA
Hoff et al ⁴⁹	2017	✓	✓	✓				TS, PR
Yang et al ⁵⁰	2019	✓	✓	✓				MA
Yaman and Elloumi ²³	2012	✓			✓			Exact Method
Sadeghi et al ²⁹	2015	✓		✓				DE
Alumur et al ³¹	2018	✓	✓	✓	✓	✓		Exact Method
Our work	2020	✓	✓	✓	✓	✓	✓	VNS, VNS-TS

on this type of problem. TS is integrated to enhance the VNS, by increasing its ability to find the global optimal.

3 | THE PROPOSED OPTIMIZATION MODEL

The proposed model is based on the CSAHLP that was developed by Ernst and Krishnamoorthy⁴⁴ with formulations of $n^3 + n^2$ variables. In flood situations, vital supplies are distributed through all villages in the affected area. To deliver the relief materials, we need to assign centers in the hub network to perform as transshipment centers. In the purposed model, total costs are associated with transportation and establishing hubs. The purpose of the network model is to establish



the sorting centers and to allocate each relief center to its hub. We consider parameters that affect transportation time and networks during a flood disaster. Different flood levels, which affect the cost function and traveling time, are considered as well as the vehicle types. Moreover, in life-saving instances, we have to be concerned about the traveling time in arcs and the total traveling time in a network. If the supplies are perishable, they will need to be delivered as soon as possible or not later than a specified time, because victims may not be able to travel far distances if the flood water is high. This model can be used to analyze different scenarios depending on the flood situation. The model consists of deterministic variables, and flood level prediction is not within the scope of this research. The details of the proposed model are described as follows.

3.1 | Model assumptions

- The number of nodes is predefined and fixed
- The number of hubs is not known
- Each non-hub node is assigned to exactly one hub
- Each hub has a limited capacity
- All hubs are fully connected
- All nodes and arcs are above the water level and operable during a flood period.
- Transportation cost and time are dependent on distance, height, water level, and type of vehicle
- Commodities are ready to be transported at each node

3.2 | Model formulation

Indexes

i, j = set of nodes; $i = \{1, 2, \dots, I\}$, $j = \{1, 2, \dots, J\}$.

k, l = set of hubs; $k = \{1, 2, \dots, K\}$, $l = \{1, 2, \dots, L\}$.

m = set of water levels; $m = \{1, 2, \dots, M\}$

t = set of transportation types; $t = \{1, 2, \dots, T\}$

Parameters

$C^{(m,t)}_{ij}$ = transportation cost from node i to node j at flood level m using vehicle type t
 $= f(d_{ij}, h_{ij}, m, t) = a^{(m,t)}(d_{ij}) + b^{(m,t)}(h_{ij})$; a and b are constant.

d_{ij} = distance from node i to node j .

h_{ij} = estimated height on the route from node i to node j .

$= \max\{\text{height at node } i, \text{height at node } j\}$

F_k = fixed cost of establishing hub k

W_{ij} = amount of flow from node i to node j

O_i = total amount of flow originating at node i , where $O_i = \sum_{j=1}^J W_{ij}$

D_j = total amount of flow to node j , where $D_j = \sum_{i=1}^I W_{ij}$

χ = discount factor for collection (non-hub to hub)

δ = discount factor for distribution (hub to non-hub)

α = discount factor for transshipment between hub links (hub to hub)



$t_{ij}^{(m,t)}$ = traveling time from node i to node j at flood level m using vehicle type t
 $= f(d_{ij}, h_{ij}, m, t) = u^{(m,t)}(d_{ij}) + v^{(m,t)}(h_{ij})$; u and v are constant.

Γ_k = maximum capacity of hub k

T_o = maximum traveling time allowed for flows from the origin traveling to the assigned hub

T_d = maximum traveling time allowed for flows from the assigned hub traveling to the destination

T_n = maximum traveling time allowed for flows traveling in the network

Decision variables

$X_{ik} = 1$ if node i is allocated to a hub located at node k ,
 $= 0$ otherwise.

$X_{kk} = 1$ if node k is a hub,
 $= 0$ otherwise.

Y_{kl}^i = total amount of flow of commodity i that is routed between hubs k and l

$$\text{Minimize } \sum_{i=1}^I k \sum_{k=1}^k C_{ik}^{(m,t)} X_{ik} (XO_i + \delta D_i) + \sum_{i=1}^I \sum_{k=1}^K \sum_{l=1}^L \alpha C_{kl}^{(m,t)} Y_{kl}^i + \sum_{k=1}^K F_k X_{kk} \quad (1)$$

Subject to

$$\sum_{k=1}^k X_{ik} = 1 \quad \forall i \quad (2)$$

$$X_{ik} \leq X_{kk} \quad \forall i, k \quad (3)$$

$$\sum_{l=1}^L Y_{kl}^i - \sum_{l=1}^L Y_{kl}^i = O_i X_{ik} - \sum_{j=1}^j W_{ij} X_{jk} \quad \forall i, k \quad (4)$$

$$\sum_{i=1}^I (O_i + D_i) X_{ik} \leq \Gamma_k X_{kk} \quad \forall k \quad (5)$$

$$\sum_{i=1}^I t_{ik}^{(m,t)} X_{ik} \leq T_o \quad \forall k \quad (6)$$

$$\sum_{j=1}^J t_{ik}^{(m,t)} X_{jk} \leq T_d \quad \forall k \quad (7)$$

$$\sum_{i=1}^I \sum_{k=1}^K t_{ik}^{(m,t)} X_{ik} + \sum_{i=1}^I \sum_{k=1}^K \sum_{l=1}^L t_{kl}^{(m,t)} Y_{kl}^i \leq T_n \quad (8)$$

$$Y_{kl}^i \geq 0 \quad \forall i, k, l \quad (9)$$

$$X_{ik} \in \{0, 1\} \quad \forall i, k \quad (10)$$

To minimize the total transportation cost, objective (1) minimizes two types of costs: the variable cost $C_{ik}^{(m,t)}$ and fixed cost F_k . The variable cost consists of three components, which are the collection cost represented by χ , the distribution cost represented by δ , and the transfer cost between hub arcs represented by α . The fixed cost is the cost of establishing hubs. Note that the collection, distribution, and transfer costs are dependent on the flood level and vehicle type. Constraint (2) ensures that each node is allocated to exactly one location.



Constraint (3) assigns the flow to a hub after the hub is located. Constraint (4) is the flow balance equation for commodity i at node k . Constraint (5) restricts the amount of arriving flow from a node to a capacitated hub. Constraints (6) and (7) limit the travel time from each node to the hubs, which depends on the flood level. Constraint (8) limits the total traveling time of the flows in the network. Constraints (9) and (10) are the non-negative and binary requirements, respectively.

4 | VNS-TS FOR SOLVING THE CSHALP

Many heuristics have been proposed for hub location problems such as the GA, TS, SA, PR, GRASP, and VNS. According to the literature, VNS-based heuristics outperform other algorithms in terms of solution quality in solving hub location problems such as the p -hub median (Ilić et al³⁹), capacitated single-allocation hub location (Marić⁵¹), p -hub maximal covering problem (Jankovic et al⁵²), p -hub center problem (Brimberg et al⁵³), and capacitated single assignment hub location problem (Mikić et al⁵⁴). Therefore, the VNS is selected to solve the proposed problem. To enhance the efficiency of the VNS, we integrate the TS to improve the solution quality. The TS strategy prevents the visitation of the allocation (node, hub) previously generated.

In this study, the VNS is developed for solving the capacitated single-allocation hub location problem. The VNS, a combinatorial optimization, was introduced by Hansen and Mladenović.⁵⁵ The VNS is based on a systematic change to avoid being trapped in local optima. In the proposed algorithm, the VNS metaheuristic contains three main steps. In the first step, the initialization defines a set of neighborhood structures N_k ($k = 1, \dots, k_{\max}$) and initial solution. In the second step, shaking randomly generates solutions from the k th neighborhoods to diversify the search and escape local optima by using different neighborhood structures. In the third step, the local search procedure improves solutions to find the local optimum.

The TS is a metaheuristic algorithm proposed by Glover⁵⁶ for solving mathematical optimizations. The main concept of the TS is building a memory structure into a local search to prevent previous solutions from being revisited for a limited period of time. A tabu list records the number of solutions that have been evaluated to avoid revisiting a solution. If an improved solution is found, a move is performed regardless of the tabu list; otherwise, moving to the new solution will only occur when the new solution is not in the tabu list. A non-improved solution is also accepted to escape the local optima. The details are presented in the following sections.

4.1 | Solution representation

In our VNS-TS algorithm, we use an array to represent a solution network. The length of the array is equal to the number of nodes, and the values in the array indicate hubs. The position in the array indicates the node, which is allocated to a hub. For instance, there are seven nodes in Figure 1, and the hubs are 4 and 6. Nodes 1, 2, and 7 are assigned to hub 4. Nodes 3 and 5 are connected to hub 6.

4.2 | Initial solution

To generate an initial solution, the algorithm performs three main steps. In the first step, we initialize a hub set. The amounts of traffic at each node ($O_i + D_i$) are calculated and sorted in

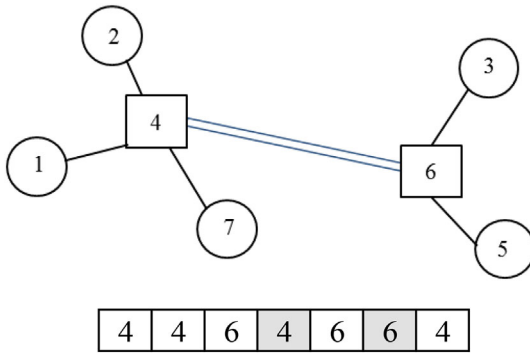


FIGURE 1 Solution representation [Color figure can be viewed at wileyonlinelibrary.com]

decreasing order. Next, we randomly predefine a number of hubs, p , and choose the initial p hubs from the non-hub nodes with the highest total flow. In the second step, we generate an initial hub network by allocating each node to its nearest hub. In the final step, the capacity restriction is checked to verify the amount of arriving flow from the node to a capacitated hub. If an infeasible solution is found, the repair function is called to transform it into a feasible solution.

4.3 | Neighborhood structures

In the proposed VNS algorithm, the neighborhood structure can be changed in three ways: (i) intra-cluster move, (ii) inter-cluster move, and (iii) allocation. In the intra-cluster move, a hub is randomly selected and replaced with a node in the same cluster. For the inter-cluster move, the hub is replaced with a node from a different cluster. Finally, for the allocation operator, a node is randomly removed from the current cluster and allocated to another hub.

4.4 | Algorithm procedure

We describe the VNS procedure in Algorithm 1 (Figure 2). The algorithm operates by selecting a set of neighborhood structures N_k , where $k = 1, \dots, k_{\max}$. Next, shaking is performed to generate a network solution x' from the k th neighborhood. After the initial hub set is assigned, each node is allocated to the nearest hub to construct the initial hub location network. The algorithm has two major parameters: (a) $T_{\max}$ represents the maximum time allowed on the search; and (b) $k_{\max}$ represents the number of neighborhoods used in shaking. In this case, $k_{\max} = 3$, and $T_{\max} = 300$ seconds or 100 iterations.

Next, we check the feasibility, which consists of the capacity and distance constraints. For the capacity restriction, we have to ensure that the hub has a sufficient capacity for traffic flows. If the capacitated constraint in a hub is violated, a few spokes in the cluster will be randomly moved to another cluster. The distance restriction contains three components: the originated distance between the non-hub node to hub, transferred distance between hubs, and destined distance between the hub and non-hub node. If a distance violation is detected, the distanced node will be randomly moved to another hub. If the violation remains, the penalty function is called to discard the solution.

To enhance the efficiency of the algorithm, we combine the VNS-TS algorithms. To prevent cycling, the recently incumbent hub set obtained from the algorithm is recorded in the tabu list.



Algorithm 1 VNS for the CSHALP

Initialisation: O_i, D_i , Neighborhood structure $N_k, (k=1, \dots, k_{max}), T_{max}$

$s_i \leftarrow s_0$

$N_i \leftarrow N_0$

while (stopping criteria is met) **do**

$k \leftarrow 1$

while ($k \leq k_{max}$) **do**

$s' \leftarrow \text{Shake}(s, k)$

$s' \leftarrow \text{Check Capacity Restriction}(s)$

$s'' \leftarrow \text{Local Search}(s')$

$k \leftarrow \text{Neighbourhood Change}(s', k)$

end while

end while

return s

FIGURE 2 VNS for the CSHALP procedure

Algorithm 2 Neighbourhood changing

If ($f(s'') < f(s')$) **then**

$s \leftarrow s''$

$k \leftarrow 1$

 Update Tabu List (s)

If ($TF > K$) **then**

$p = p + 1$

else

$p = p - 1$

end if

else

$k \leftarrow k + 1$

end if

FIGURE 3 Neighborhood changing procedure

To prevent the same movement in a tabu tenure length, a tabu list is obtained by recording the best allocation pair (node, hub). Through a preliminary examination, tabu tenure lengths of 4-9 are examined. The best solution quality is obtained from a tabu tenure length of 7. In the local search, we employ neighborhood changing functions as described in Algorithm 2. To diversify the solutions, we use three neighborhood structures, which are an intra-cluster move, inter-cluster move, and allocation. Then, we calculate the total transportation and fixed costs of the network (TF) to adjust number of hubs (p). If the total transportation cost and the fixed cost ratio is higher than K , we increase p by 1. In the same way, p will be decreased by 1 if the total transportation cost and the fixed cost ratio is less than K . The neighborhood changing algorithm details are shown in Algorithm 2 (Figure 3).



To measure the solution quality and control the computational time, the gap is defined as the percentage above the optimal solution when optimal solution is known, or the best known solution when the optimal solution is not known as in Equation (11). Note that here we consider the minimization problem, and a lower solution is preferred.

$$\%Gap = \left(\frac{\text{Incumbent}}{\text{BestKnown (or Optimal Sol)}} - 1 \right) * 100 \quad (11)$$

4.5 | Stopping criteria

When the optimal solutions are known, the program will be terminated if the search finds an incumbent within the setting value, which is set at $\%gap = 0.5$. The number of objective function evaluations is used as a stopping criterion if the gap is outside the setting value. If this is the case, we allow up to 100 iterations. These values were chosen to allow the running times to be reasonable.

5 | COMPUTATIONAL EXPERIMENTS

In this section, we test the performance of the hybrid VNS-TS algorithm through a comparison between the VNS algorithm and the optimization program language (OPL). The proposed hybrid algorithm is first tested on solving the CSAHLP using an open standard dataset. Note that the CSAHLP is the proposed model without the distance restricted constraints and with one dimension of cost and capacity. The AP dataset is a set of benchmark data that was introduced by Ernst and Krishnamoorthy.³⁶ The data selected for this study consist of 16 subproblems with combinations of four different problem sizes (10, 20, 40, and 50 nodes) and four different types of assigned fixed cost (X) and capacity (Y), represented by XY, where $X, Y \in \{T, L\}$; T = Tight, L = Loose, and T indicates that the problem is harder to solve than is L (LL, LT, TL, TT). Both algorithms are coded in C Programming using the Mersenne Twister Random number generator. All computational experiments are performed on an Intel Core 2.4 GHz with 2 GB RAM computer. The results are shown in Table 2. We evaluate the solution quality based on the solving time (CPU time) in seconds and the gap between a current solution and the optimal solution ($\%gap$). Note that in this case, we use the standard dataset; therefore, the optimal solutions are known in all subproblems. Both algorithms (VNS and VNS-TS) found the solution within a few seconds: an average of 4.93 seconds for the VNS and an average of 11.60 seconds for the VNS-TS; however, the OPL took longer to find the solution, an average of 837.77 seconds. If we look at the solution quality, the VNS-TS performs better than VNS and OPL with a 0% gap in all subproblems, while the VNS found optimal solutions in 11 out of 16 subproblems (68.75%), and the OPL found optimal solutions in 15 out of 16 subproblems (93.75%). Note that we terminated the OPL after 3.5 hours.

6 | CASE STUDY OF SEVERE FLOODING IN THAILAND

Our case study is from the Nonthaburi Province, which is located in the central region of Thailand. The province covers an area of 622 km² with a population of 1.2 million. The administrative area consists of six districts, which are Muang, Bang-kruai, Bang-yai,

**TABLE 2** Computational results with the AP dataset

Problem	Optimal cost	OPL		VNS		VNS-TS	
		% Gap	CPU time (s)	% Gap	CPU time (s)	% Gap	CPU time (s)
10LL	224 250.05	0.00	8.86	0.00	0.029	0.00	0.05
10LT	250 992.26	0.00	6.56	0.00	0.03	0.00	0.08
10TL	263 399.94	0.00	17.15	0.00	0.03	0.00	0.50
10TT	263 399.94	0.00	7.35	0.00	0.03	0.00	0.49
20LL	234 690.96	0.00	6.08	0.00	0.40	0.00	0.46
20LT	253 517.40	0.00	7.01	0.00	0.50	0.00	0.39
20TL	271 128.18	0.00	5.58	0.00	0.63	0.00	0.72
20TT	296 035.40	0.00	8.63	0.82	0.31	0.00	12.37
40LL	241 955.71	0.00	24.60	0.00	3.87	0.00	19.80
40LT	272 218.32	0.00	31.92	0.00	5.63	0.00	12.21
40TL	298 919.01	0.00	19.46	0.00	5.09	0.00	15.23
40TT	354 874.10	0.00	112.85	0.45	6.34	0.00	18.76
50LL	238 520.59	0.00	35.02	0.00	9.03	0.00	19.59
50LT	272 897.49	0.00	223.70	0.69	18.55	0.00	22.05
50TL	319 015.77	0.00	145.72	0.36	9.28	0.00	28.96
50TT	417 440.99	0.000046	12 743.89 ^a	0.94	19.07	0.00	34.06
Average		0.0000028	837.77	0.20	4.93	0.00	11.60

^aCPU time that we found the best known for OPL.

Bang-buatong, Sai-noi, and Pak-kret. Nonthaburi is one of the critical regions affected by severe flooding in Thailand in 2011. Without experience and improper preparedness, several people in the affected area suffered from a lack of water, food, and survival supplies. Establishing relief centers at appropriate locations helps us to more efficiently organize the delivery of supplies to victims. By considering the geographical factors, 68 existing locations (nodes) are selected to be candidates for hubs,⁵⁷ assuming that demand is generated at the center of each node. The map of Nonthaburi Province and the locations of the hub are shown in Figure 4.

We conduct the experiments by classifying the floodwater into two levels: low at 0.5–1.0 m and high at 1.0–1.5 m. Note that at different water levels, we need different types of vehicles for transportation. The transportation cost and time are dependent on the water level, type of vehicle, and height of the area. High water levels cause higher transportation costs and times. The hub capacities are also considered to be loose or tight. The discount factors for transshipments between hub links (α) are varied: 0.4, 0.6, and 0.8. The discount factors for collection and distribution are fixed at 3 and 2, respectively, based on Ernst and Krishnamoorthy.⁴⁴ Then, we solve the problem based on three scenarios: (1) without a bound on the travel time, (2) with a bound on travel time at 40 minutes, and (3) with a bound on travel time at 20 minutes, for a total of 36 cases. We assumed that vehicles travel at a speed of 1 km/min.

We found an appropriate number of hubs, their locations, and the routes from demand nodes to hubs by solving the proposed model, with the dataset of our case study as the input to the model.

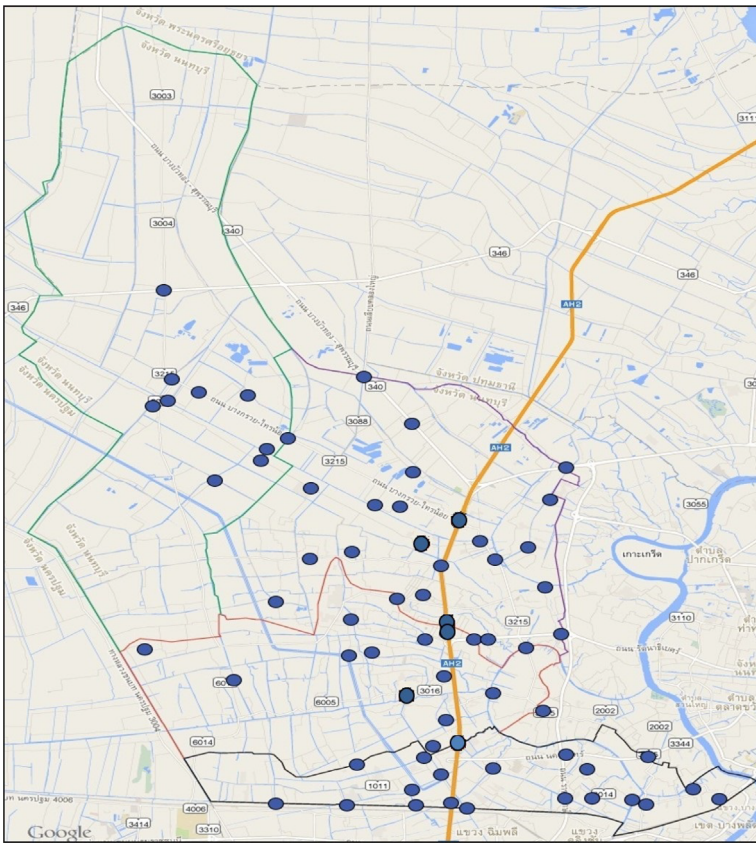


FIGURE 4 Map of candidate hubs in Nonthaburi Province [Color figure can be viewed at wileyonlinelibrary.com]

The problem was solved by optimization software and two metaheuristic algorithms: the VNS and VNS-TS. Note that the problem in Scenario 1 consisted of 9384 constraints and 319 056 variables: 4624 binary variables and 314 432 other variables. The problems in Scenarios 2 and 3 consisted of 14 008 constraints and 319 056 variables: 4624 binary variables and 314 432 other variables. The computational results are shown in Tables 3–8. We report the best known and hub sets for Scenarios 1, 2, and 3 in Tables 3, 5, and 7. The solution quality (%gap) and running time (s) of Scenarios 1, 2, and 3 are reported in Tables 4, 6, and 8. The hub set of the first scenario shows that with no bound on the arc, a fewer number of hubs are opened to minimize the total transportation cost. This result may not be practical, because people have to travel long distances. In the second and third scenarios, larger numbers of hubs are opened, especially when capacity is tight, and people do not have to travel long distances during a disaster. Based on these experiments, setting the bound at 20 minutes is preferable to the victims. However, there will be trade-offs between the service and the increasing transportation cost.

The OPL required more than an hour in some cases to find the optimal solution. The average running times of the three scenarios were 22.40, 19.70, and 6.71 minutes, respectively. Meanwhile, the average running times of the VNS and VNS-TS were approximately 1 minute. In the first scenario, without conditions on the travel time, the VNS-TS found the best known solution (with 0% gap) in an average of 49.78 seconds, while the VNS did the best at a 0–0.8% gap with an average of 25.18 seconds. In the second scenario, with the travel time limited to 40 minutes, the VNS-TS found the best known solution (with 0% gap) in an average of 113.10 seconds, while the VNS did the best at the 0–2% gap with an average of 63.23 seconds. In the third scenario, with a limit on

TABLE 3 Best known and hub set for Scenario 1 without a limit on the travel time

				OPL	
Flood level	Alpha	Hub capacity	Best known	CPUtime (min)	Hub set
Low	0.4	Loose	568 948.92	4.26	{14 30 36}
		Tight	757 983.92	12.13	{14 24 30 36 62}
	0.6	Loose	582 837.68	0.50	{14 30 36}
		Tight	772 007.01	8.35	{14 30 36}
	0.8	Loose	596 713.21	12.35	{14 30 36}
		Tight	786 030.25	15.41	{14 24 30 36 62}
High	0.4	Loose	1 180 803.90	21.49	{14 36 63}
		Tight	1 439 282.19	14.16	{4 14 24 36 63}
	0.6	Loose	1 218 034.83	74.23	{14 36 63}
		Tight	1 485 830.66	15.15	{4 14 24 36 63}
	0.8	Loose	1 255 265.75	60.20	{14 36 63}
		Tight	1 532 258.21	30.51	{4 14 24 36 63}
Average				22.40	

TABLE 4 Gap and solving time for Scenario 1 without a limit on the travel time

				VNS	VNS-TS		
Flood level	Alpha	Hub capacity	Best known	%Gap	CPU time (s)	%Gap	CPU time (s)
Low	0.4	Loose	568 948.92	0.00	17.43	0.00	32.09
		Tight	757 983.92	0.00	23.90	0.00	42.89
	0.6	Loose	582 837.68	0.00	19.96	0.00	47.99
		Tight	772 007.01	0.81	28.07	0.00	56.30
	0.8	Loose	596 713.21	0.00	18.00	0.00	42.92
		Tight	786 030.25	0.00	29.39	0.00	61.31
High	0.4	Loose	1 180 803.90	0.00	30.92	0.00	57.47
		Tight	1 439 282.19	0.00	28.79	0.00	49.04
	0.6	Loose	1 218 034.83	0.00	23.99	0.00	39.67
		Tight	1 485 830.66	0.00	28.84	0.00	67.42
	0.8	Loose	1 255 265.75	0.00	23.93	0.00	50.39
		Tight	1 532 258.21	0.00	28.89	0.00	49.88
Average				0.07	25.18	0.00	49.78

TABLE 5 Best known and hub set for Scenario 2 with a limit on travel time (40 minutes)

				OPL	
Flood level	Alpha	Hub capacity	Best known	CPUtime (m)	Hub set
Low	0.4	Loose	582 837.68	10.43	{14 30 36}
		Tight	772 007.00	12.05	{14 24 30 36 62}
	0.6	Loose	568 948.92	11.10	{14 30 36}
		Tight	757 983.76	9.51	{14 24 30 36 62}
	0.8	Loose	596 713.21	11.48	{14 30 36}
		Tight	786 030.25	12.29	{14 24 30 36 62}
High	0.4	Loose	1 184 709.65	48.21	{14 36 63}
		Tight	1 184 709.65	9.16	{4 14 24 36 63}
	0.6	Loose	1 221 953.71	45.32	{14 36 63}
		Tight	1 485 814.93	16.18	{4 14 24 36 63}
	0.8	Loose	1 259 145.38	26.12	{14 36 63}
		Tight	1 532 258.21	24.55	{4 14 24 36 63}
Average				19.70	

TABLE 6 Gap and solving time for Scenario 2 with a limit on the travel time (40 minutes)

				VNS		VNS-TS	
Flood level	Alpha	Hub capacity	Best known	%Gap	CPU time (s)	%Gap	CPU time (s)
Low	0.4	Loose	1 184 709.65	0.00	57.17	0.00	92.35
		Tight	1 184 709.65	0.00	90.75	0.00	200.49
	0.6	Loose	1 221 953.71	0.00	44.53	0.00	84.61
		Tight	1 485 814.93	0.00	78.67	0.00	116.29
	0.8	Loose	1 259 145.38	0.00	48.94	0.00	72.65
		Tight	1 532 258.21	0.80	68.27	0.00	91.57
High	0.4	Loose	1 184 709.65	0.00	56.68	0.00	102.68
		Tight	1 184 709.65	1.32	57.72	0.00	97.75
	0.6	Loose	1 221 953.71	0.17	50.25	0.00	87.30
		Tight	1 485 814.93	0.78	84.88	0.00	148.32
	0.8	Loose	1 259 145.38	0.21	52.81	0.00	139.88
		Tight	1 532 258.21	2.28	68.08	0.00	123.35
	Average			0.46	63.23	0.00	113.10



TABLE 7 Best known and hub set for Scenario 3 with a limit on the travel time (20 minutes)

Flood level	Alpha	Hub capacity	Best known	OPL	
				CPUtime (m)	Hub set
Low	0.4	Loose	568 948.92	5.39	{14 30 36}
		Tight	757 983.76	14.39	{14 24 30 36 62}
	0.6	Loose	582 837.68	8.35	{14 30 36}
		Tight	772 007.01	19.45	{14 24 30 36 62}
	0.8	Loose	596 713.21	14.45	{14 30 36}
		Tight	786 030.25	9.17	{14 24 30 36 62}
High	0.4	Loose	3 238 775.57	1.04	{1 3 14 16 20 22 34 37 38 42 44 45 47 53 63}
		Tight	3 587 448.61	1.35	{3 6 14 16 20 21 22 34 37 38 42 44 45 47 53 63 65}
	0.6	Loose	3 284 836.91	1.08	{1 3 14 16 20 22 34 37 38 42 44 45 47 53 63}
		Tight	3 635 536.63	2.10	{3 6 14 16 20 21 22 34 37 38 42 44 45 47 53 63 65}
	0.8	Loose	3 330 898.25	1.35	{1 3 14 16 20 22 34 37 38 42 44 45 47 53 63}
		Tight	3 683 586.38	2.38	{3 6 14 16 20 21 22 34 37 38 42 44 45 47 53 63 65}
	Average			6.41	

travel time of 20 minutes, the VNS-TS found the best known solution (with 0% gap) in an average of 103.05 seconds, while the VNS did the best at a 0-0.9% gap with an average of 59.83 seconds. For solving real problems, we focus more on the quality of the solutions, although the solving times of both algorithms are different. Therefore, in this case the VNS-TS is preferred over the VNS.

To compare the performances of the VNS and VNS-TS, we performed a hypothesis test on the difference between these two algorithms, using the %gap and runtime for all the scenarios in Tables 4, 6, and 8. At a significance level of 0.05, there is a significant difference between the VNS gap percentage and VNS-TS gap percentage, with the *P*-value equal to .005714 (the average %gap of VNS = 0.24, the average %gap of VNS-TS = 0.00). There is also a significant difference between the VNS runtime and VNS-TS runtime with a *P*-value of .000003 (the average VNS run time = 49.41 seconds, the average VNS-TS run time = 88.65 seconds). Based on these results, the VNS-TS performed better than did the VNS in terms of solution quality; however, it requires a longer solving time.

7 | SENSITIVITY ANALYSIS

In this section, we perform a sensitivity analysis on the involved parameters, which are the number of flows, discount factor, hub capacity, and bound length, to examine their impacts on the total transportation cost, number of hubs, and hub locations.

**TABLE 8** Gap and solving time for Scenario 3 with a limit on the travel time (20 minutes)

				VNS		VNS-TS	
Flood level	Alpha	Hub capacity	Best known	%Gap	CPU time (s)	%Gap	CPU time (s)
Low	0.4	Loose	568 948.92	0.00	51.77	0.00	74.68
		Tight	757 983.76	0.00	90.95	0.00	186.49
	0.6	Loose	582 837.68	0.00	40.93	0.00	65.41
		Tight	772 007.01	0.00	74.27	0.00	110.69
	0.8	Loose	596 713.21	0.00	47.14	0.00	69.45
		Tight	786 030.25	0.67	60.07	0.00	88.37
High	0.4	Loose	3 238 775.57	0.00	49.48	0.00	83.48
		Tight	3 587 448.61	0.50	55.12	0.00	84.15
	0.6	Loose	3 284 836.91	0.00	49.45	0.00	81.70
		Tight	3 635 536.63	0.20	81.68	0.00	142.32
	0.8	Loose	3 330 898.25	0.00	53.01	0.00	124.68
		Tight	3 683 586.38	0.93	9.84	0.00	125.20
Average				0.19	59.83	0.00	103.05

In this case, we consider a variation in the commodities or flows in the network. In reality, it is difficult to estimate the number of flows during a disaster. We consider the case that is the hardest to manage, which is when the water level is high and the number of flows changes. We vary the number of flows (λ) from 0.8 to 2.0. Note that $\lambda = 1.0$ refers to the normal case as presented in the previous section, $\lambda = 0.8$ refers to the case in which there are 20% fewer flows than expected, and $\lambda = 1.2$ refers to the case in which there are 20% more flows than expected. The discount factor also varies and is equal to 0.4, 0.6, and 0.8, and the hub capacity is either loose or tight. The results are shown in Table 9.

Figures 5 and 6 show the total transportation cost when the capacities are large (loose) and limited (tight). The total transportation cost increases when the number of flows increases and is higher with a limited capacity. For the discount factors, the total transportation cost increases as the transshipment discount factor increases. The impact is greater with a high number of flows.

Figures 7 and 8 show the number of hubs with large and limited capacities. The number of hubs and hub sets do not change much with a large capacity, between 15 and 20 hubs. However, when the capacity is limited, we have to open a greater number of hubs, between 17 and 34 hubs. If the hub set changes, the route pattern of flows also changes.

Next, we vary the transshipment discount factor from 0.1 to 3 to examine the impact on the total transportation cost, which is composed of variable and fixed costs. The collection and distribution discount factors are set at 3 and 2, respectively. The number of flows is set at the normal case ($\lambda = 1.0$), and the bound length is set at 20 minutes. Figure 9 shows the differences between the total transportation costs for loose and tight capacities when varying the discount factors. In Figures 10 and 11, we break down the total cost into variable and fixed costs for both the loose and tight capacity cases. The total transportation cost is higher when the capacity is tight. However, the fixed cost is higher than the variable cost for all cases. In particular, the fixed cost remains the same for all cases, while the variable cost increases as the discount factor increases. Note that

TABLE 9 Comparison of the total transportation cost and hub sets for different numbers of flows

Number of flows	Alpha	Hub capacity	Total cost	Hub set
0.8	0.4	Loose	3 139 898.12	{1 3 14 16 20 22 34 37 38 42 44 45 47 53 63}
		Tight	3 381 017.90	{3 6 14 16 20 21 22 34 37 38 42 44 45 47 53 62 63}
	0.6	Loose	3 176 747.19	{1 3 14 16 20 22 34 37 38 42 44 45 47 53 63}
		Tight	3 418 518.40	{3 6 14 16 20 21 22 34 37 38 42 44 45 47 53 62 63}
	0.8	Loose	3 213 596.27	{1 3 14 16 20 22 34 37 38 42 44 45 47 53 63}
		Tight	3 455 864.49	{3 6 14 16 20 21 22 34 37 38 42 44 45 47 53 62 63}
1	0.4	Loose	3 238 775.57	{1 3 14 16 20 22 34 37 38 42 44 45 47 53 63}
		Tight	3 587 448.61	{3 6 14 16 20 21 22 34 37 38 42 44 45 47 53 63 65}
	0.6	Loose	3 284 836.91	{1 3 14 16 20 22 34 37 38 42 44 45 47 53 63}
		Tight	3 635 536.63	{3 6 14 16 20 21 22 34 37 38 42 44 45 47 53 63 65}
	0.8	Loose	3 330 898.25	{1 3 14 16 20 22 34 37 38 42 44 45 47 53 63}
		Tight	3 683 586.38	{3 6 14 16 20 21 22 34 37 38 42 44 45 47 53 63 65}
1.2	0.4	Loose	4 298 358.11	{3 6 14 16 20 21 22 34 37 38 42 44 45 47 53 63 65}
		Tight	5 119 754.78	{3 6 8 14 17 20 21 22 25 34 36 37 38 42 44 45 47 51 53 62 63 65 66}
	0.6	Loose	4 414 684.58	{3 6 14 16 20 21 22 34 37 38 42 44 45 47 53 63 65}
		Tight	5 239 254.96	{3 6 8 14 17 20 21 22 25 34 36 37 38 42 44 45 47 51 53 62 63 65 66}
	0.8	Loose	4 530 989.10	{3 6 14 16 20 21 22 34 37 38 42 44 45 47 53 63 65}
		Tight	5 358 755.14	{3 6 8 14 17 20 21 22 25 34 36 37 38 42 44 45 47 51 53 62 63 65 66}
1.4	0.4	Loose	4 607 926.69	{3 6 14 16 20 21 22 34 37 38 42 44 45 47 53 62 63 65}
		Tight	5 723 267.86	{3 6 8 14 17 20 21 22 24 25 27 30 34 36 37 38 39 42 44 45 47 51 53 62 63 65}
	0.6	Loose	4 744 397.56	{3 6 14 16 20 21 22 34 37 38 42 44 45 47 53 62 63 65}
		Tight	5 864 205.40	{3 6 8 14 17 20 21 22 24 25 27 30 34 36 37 38 39 42 44 45 47 51 53 62 63 65}
	0.8	Loose	4 880 729.51	{3 6 14 16 20 21 22 34 37 38 42 44 45 47 53 62 63 65}
		Tight	6 004 739.75	{3 6 8 14 17 20 21 22 24 25 27 30 34 36 37 38 39 42 44 45 47 51 53 62 63 65}

(Continues)



TABLE 9 (Continued)

Number of flows	Alpha	Hub capacity	Total cost	Hub set
1.6	0.4	Loose	4 918 594.80	{3 6 14 16 20 21 22 25 34 37 38 42 44 45 47 48 53 63 65}
		Tight	6 699 864.37	{3 5 6 11 17 20 21 22 24 27 30 34 36 37 38 39 42 44 45 46 53 54 58 62 63 64 65}
	0.6	Loose	5 077 452.44	{3 6 14 16 20 21 22 25 34 37 38 42 44 45 47 48 53 63 65}
		Tight	6 864 923.88	{3 5 6 11 17 20 21 22 24 27 30 34 36 37 38 39 42 44 45 46 53 54 58 62 63 64 65}
	0.8	Loose	5 235 822.61	{3 6 14 16 20 21 22 25 34 37 38 42 44 45 47 48 53 63 65}
		Tight	7 029 812.87	{3 5 6 11 17 20 21 22 24 27 30 34 36 37 38 39 42 44 45 46 48 50 53 58 62 63 64 65}
1.8	0.4	Loose	5 185 436.91	{3 6 14 16 20 21 22 25 34 37 38 42 44 45 47 53 62 63 65}
		Tight	7 389 045.53	{2 3 6 11 16 17 20 21 23 24 26 27 30 34 35 36 37 38 40 42 44 45 46 48 49 51 53 54 58 62 63 65}
	0.6	Loose	5 362 580.86	{3 6 14 16 20 21 22 25 34 37 38 42 44 45 47 53 62 63 65}
		Tight	7 579 695.13	{2 3 6 11 16 17 20 21 23 24 26 27 30 34 35 36 37 38 40 42 44 45 46 48 49 51 53 54 58 62 63 65}
	0.8	Loose	5 539 431.08	{3 6 14 16 20 21 22 25 34 37 38 42 44 45 47 53 62 63 65}
		Tight	7 770 344.72	{2 3 6 11 16 17 20 21 23 24 26 27 30 34 35 36 37 38 40 42 44 45 46 48 49 51 53 54 58 62 63 65}
2.0	0.4	Loose	5 536 671.35	{3 6 14 17 20 21 22 25 34 37 38 42 44 45 47 51 53 62 63 65}
		Tight	7 978 625.13	{3 5 6 8 14 16 17 20 21 22 23 24 27 28 30 34 35 36 38 39 40 42 44 45 46 48 49 53 54 58 61 62 63 65}
	0.6	Loose	5 735 101.59	{3 6 14 17 20 21 22 25 34 37 38 42 44 45 47 51 53 62 63 65}
		Tight	8 189 783.46	{3 5 6 8 14 16 17 20 21 22 23 24 27 28 30 34 35 36 38 39 40 42 44 45 46 48 49 53 54 58 61 62 63 65}
	0.8	Loose	5 933 305.69	{3 6 14 17 20 21 22 25 34 37 38 42 44 45 47 51 53 62 63 65}
		Tight	8 400 941.78	{3 5 6 8 14 16 17 20 21 22 23 24 27 28 30 34 35 36 38 39 40 42 44 45 46 48 49 53 54 58 61 62 63 65}

solutions with the same fixed cost have the same hub set. Therefore, based on these results, the transshipment discount factor only impacts the variable cost.

To see the impact of the hub capacity, we varied the levels of hub capacity to be 0.25, 0.5, 1, 2, and 5. Note that a capacity level of 1 is the case of loose capacity, and the capacity level of 2 is twice the capacity of the loose case. The number of flows was set at normal ($\lambda = 1.0$), the bound length was set at 20 minutes, and the discount factor varied at values equal to 0.4, 0.6, and 0.8. Figure 12 shows that the total transportation cost decreases as the hub capacity increases, when the capacity level is 0.25-1. However, increasing the capacity level to more than 1 does not improve the total

FIGURE 5 Total transportation cost when varying the number of flows with a loose capacity[Color figure can be viewed at [wileyonlinelibrary.com](#)]

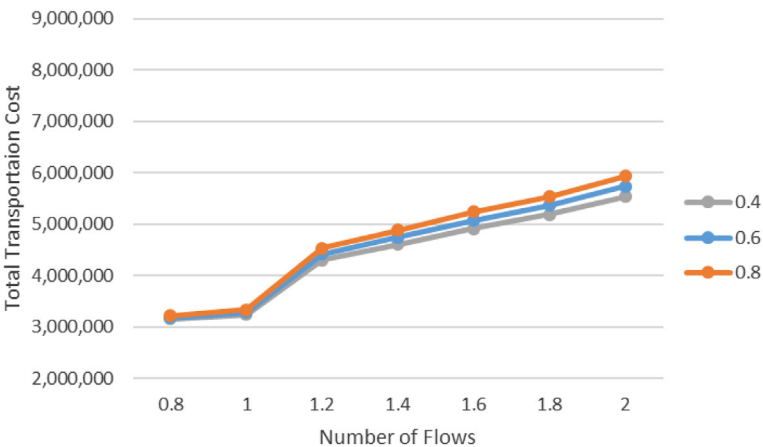


FIGURE 6 Total transportation cost when varying the number of flows with a tight capacity[Color figure can be viewed at [wileyonlinelibrary.com](#)]

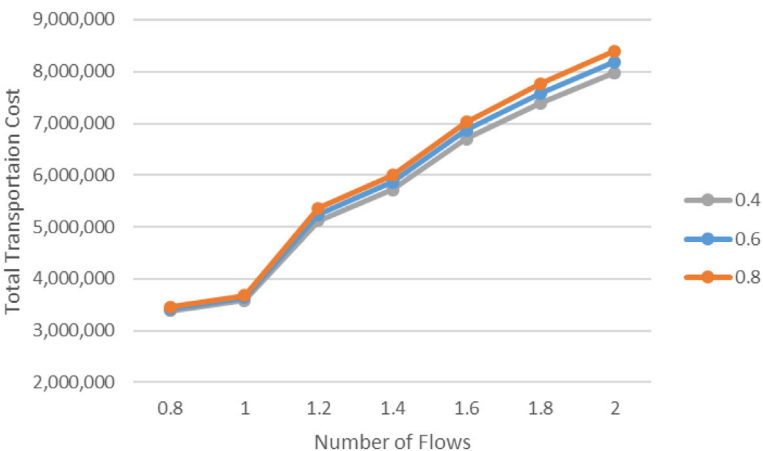
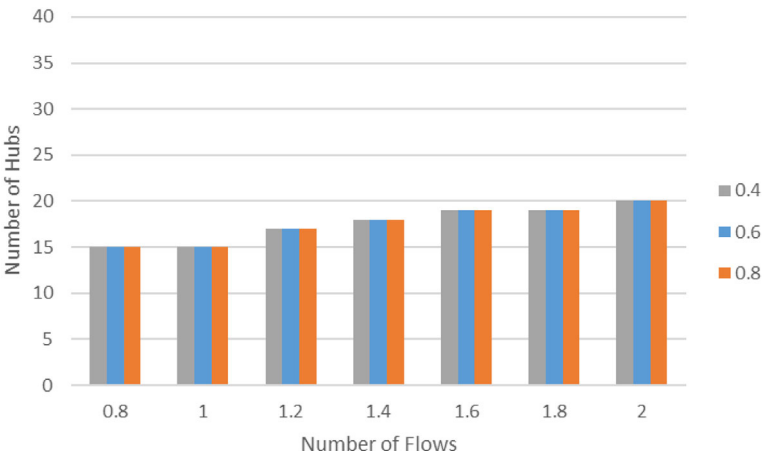


FIGURE 7 Number of hubs when varying the number of flows with a loose capacity[Color figure can be viewed at [wileyonlinelibrary.com](#)]



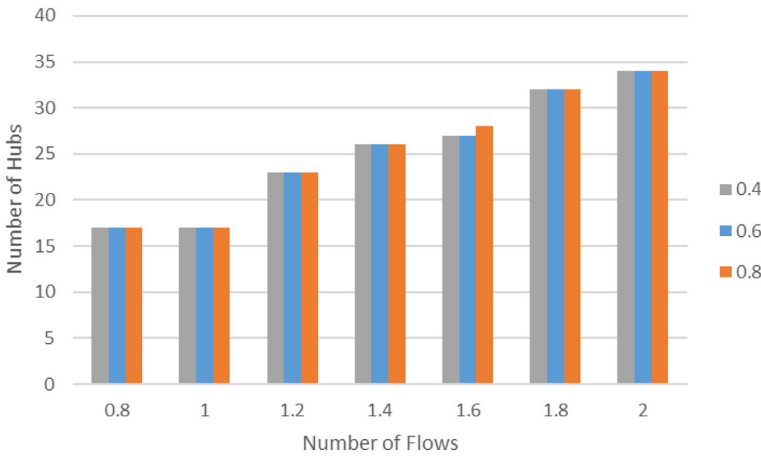


FIGURE 8 Number of hubs when varying the number of flows with a tight capacity[Color figure can be viewed at wileyonlinelibrary.com]

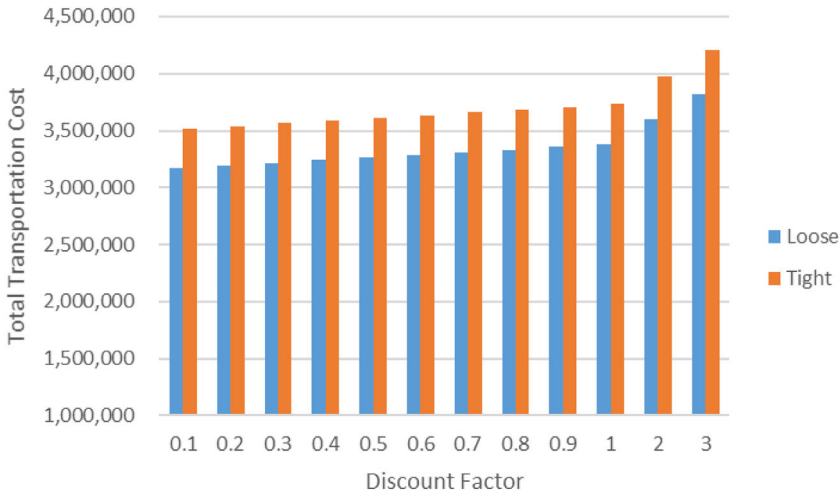


FIGURE 9 Total transportation cost when varying the discount factor with different capacities[Color figure can be viewed at wileyonlinelibrary.com]

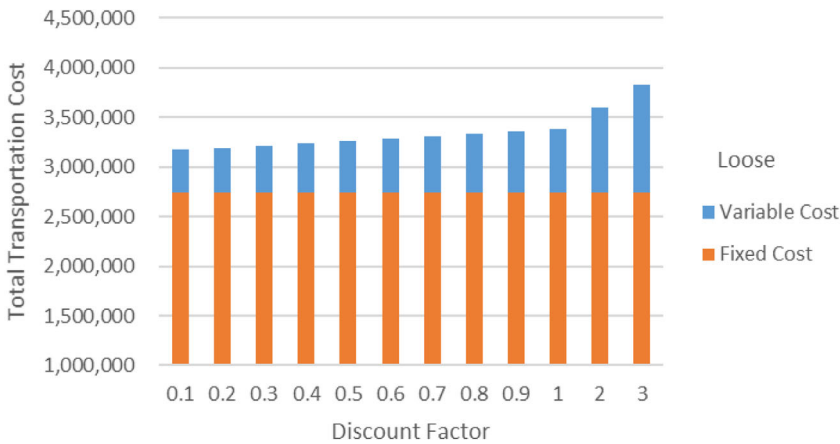


FIGURE 10 Comparison of different costs when varying the discount factor with a loose capacity[Color figure can be viewed at wileyonlinelibrary.com]

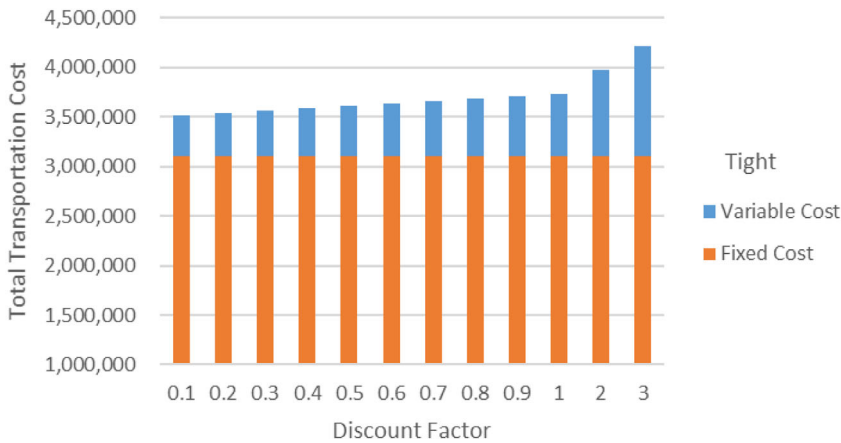
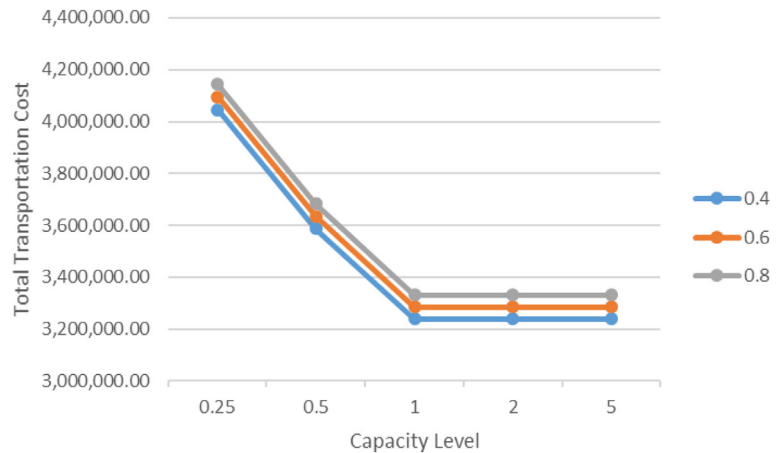


FIGURE 11 Comparison of different costs when varying the discount factor with a tight capacity[Color figure can be viewed at wileyonlinelibrary.com]

FIGURE 12 Total transportation cost with varying capacity levels[Color figure can be viewed at wileyonlinelibrary.com]



transportation cost. The solution is prohibited because of the set bound length. Therefore, in the next section, we further investigate the impact of the bound length.

Based on the previous results, no solutions exist if a short bound length is set. Only a few solutions are feasible if a short travel time exists. Therefore, in the following test, we vary the bound length to values of 20, 40, 60, 80, 100, and 200 minutes to examine the impact on the solutions. The number of flows is set at the normal case ($\lambda = 1.0$), the capacity level is fixed at 1, and the discount factor is set at 0.4, 0.6, and 0.8. The results are shown graphically in Figures 13 and 14. It can be seen that the total transportation cost is significantly higher when the bound length is 20 minutes. In this case, 15 hubs are open to allow the flows that can be shipped in the bound length period. In other cases, when we relax the bound length period, the total transportation cost can be reduced by opening a fewer number of hubs. For the cases of bound length between 40 and 100 minutes, three hubs are open. When the bound length is 200 minutes, two hubs are open at a discount factor of 0.4, while three hubs are open at discount factors of 0.6 and 0.8. As there is a big gap in the solutions between bound lengths of 20 and 40 minutes, we look for more solutions in this length. The total transportation cost and number of hubs are reported graphically

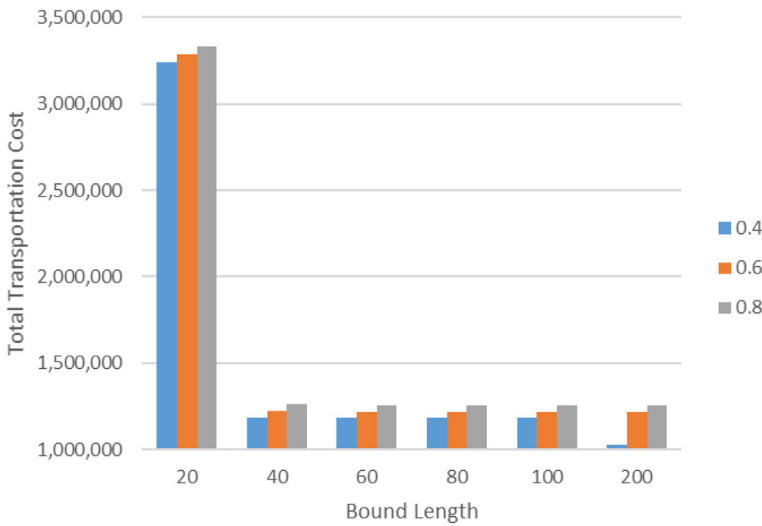


FIGURE 13 Total transportation cost when varying the bound length between 20 and 200 minutes[Color figure can be viewed at wileyonlinelibrary.com]

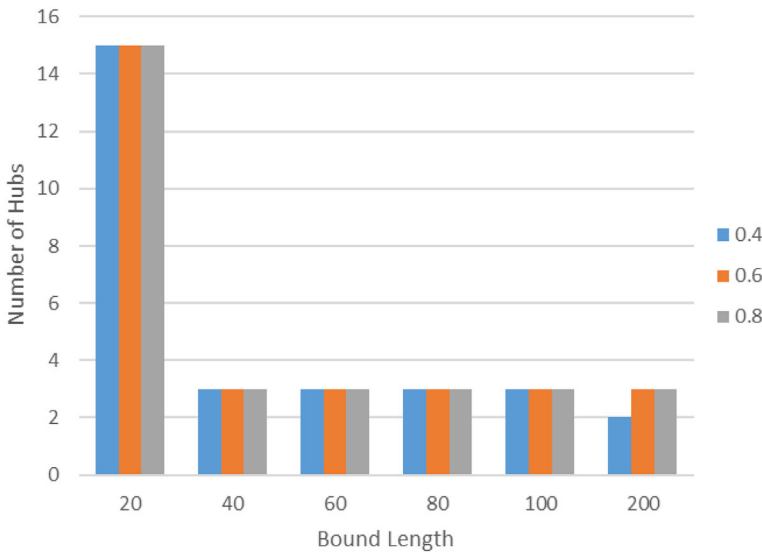


FIGURE 14 Number of hubs when varying the bound length between 20 and 200 minutes[Color figure can be viewed at wileyonlinelibrary.com]

in Figures 15 and 16. It can be seen that the total transportation cost can be reduced by increasing the bound length. The number of hubs also decreases as the bound length increases. Opening more hubs leads to a higher total transportation cost. In this case, we have various solutions, which is helpful for a decision maker to provide alternatives for hub network planning.

8 | CONCLUSION AND FUTURE WORK

This article proposes an optimization model for a transportation network during a flood disaster by locating hubs as relief centers and assigning nodes as origins of supplies or destinations of needs. The objective is to minimize total transportation costs. Several factors under consideration include water levels, center capacities, and time periods to receive supplies. Owing



FIGURE 15 Total transportation cost when varying the bound length between 20 and 40 minutes at $\alpha = 0.4$ [Color figure can be viewed at wileyonlinelibrary.com]

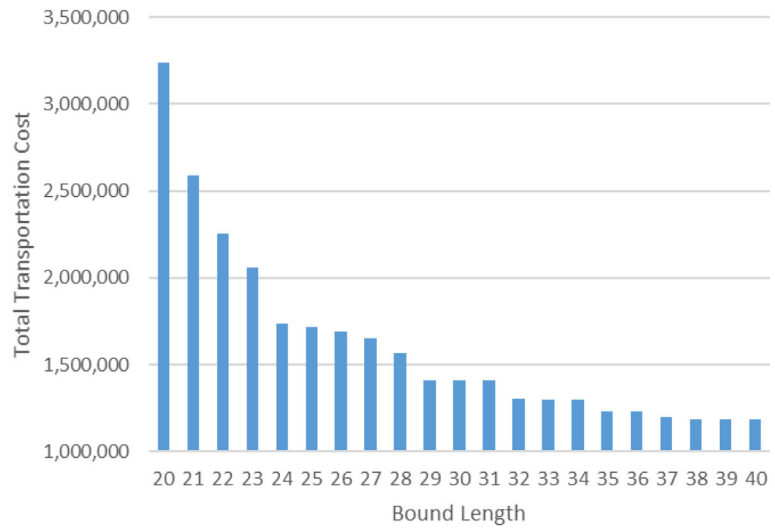
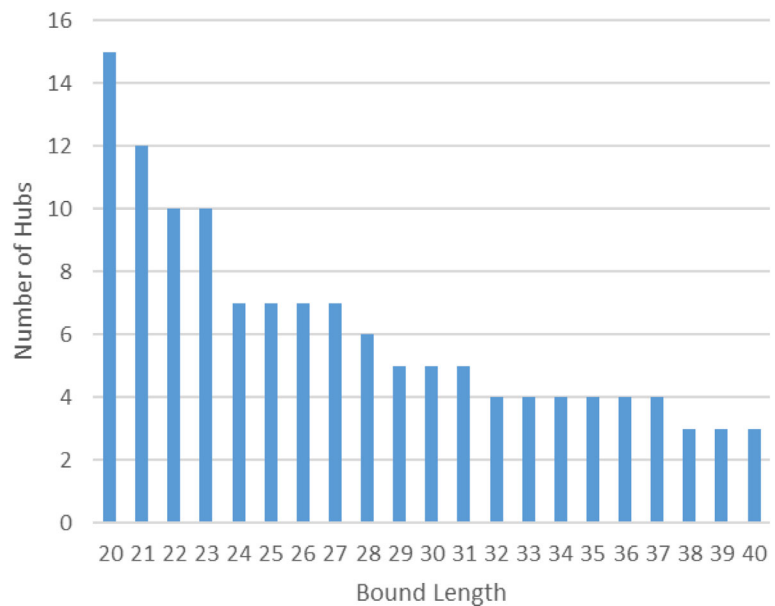


FIGURE 16 Number of hubs when varying the bound length between 20 and 40 minutes at $\alpha = 0.4$ [Color figure can be viewed at wileyonlinelibrary.com]



to the complexity of the problem, a hybrid VNS-TS algorithm is chosen as the solution method. The performance of the proposed algorithm is tested on the benchmark problem with the AP dataset and compared with those of the VNS and OPL. For most of the test problems, both the VNS and VNS-TS reach optimal solutions within acceptable computational times. The OPL requires more time to find optimal solutions in most cases and experiences difficulty in finding the optimal solution for large-sized problems. The VNS reaches the solution faster for large-sized problems; however, the VNS-TS yields better results in term of solution quality with a 0% gap in all of the cases tested.

The proposed algorithm is also applied to a real-world case study of severe flooding in Thailand in 2011. Computational experiments are conducted by considering parameters including the flood water levels, low and high; capacities of hubs, loose and tight; and discount factors. The experiments consist of three scenarios: (1) without a bound on the travel time, (2) with a bounded



travel time of 40 minutes, and (3) with a bounded travel time of 20 minutes. The results show that in the case without a limit on the travel time, the solution tends to open a fewer number of hubs to minimize the total transportation cost. In contrast, in the case with a limit on the travel time, the solution tends to open more hubs, especially when their capacity is limited. The OPL and VNS-TS find all of the optimal solutions, while the VNS finds an optimal solution with a 2% gap. Therefore, the VNS-TS yields a better solution in term of quality; however, it requires more solving time with a statistically significant difference at a significance level of 0.05. The sensitivity analysis also shows the impact of the number of flows on the total transportation cost, number of hubs, and route pattern, especially with a limited capacity and short bound length. Other factors are also analyzed, such as discount factors, capacity level, and bound length.

The most important procedure of both the VNS and VNS-TS algorithms is providing the best hub first and then accommodating the best allocation. Finding the feasible hubs takes a shorter time compared with allocating arcs to nodes. This enables the VNS to find the solution slightly faster than does the VNS-TS, because with the VNS-TS, more solutions need to be evaluated, and the reallocating procedure is performed to find the best arc. To investigate the optimal solution, the VNS-TS applies the concept that there are some common characteristics with local optima and an optimal solution. The VNS-TS intensifies the search space for the optimal hub set in the elite pool of local optima obtained from the VNS. That is, using the VNS-TS algorithm as a hybridization can improve the performance of the construction algorithm.

In this study, we assume that all nodes and arcs are above the water level and are able to operate during a flood period. In reality, some nodes or arcs may be disrupted by high flood levels. We also assume that there is only one type of commodity with the same conditions on the delivery time. As a direction for future research, more complex hub location problems with real-world parameters would be interesting to study including the risk of breakdown or disruption affecting a transportation network. The multicommodity problem is also an important issue, as different types of supplies require different conditions on the delivery time. For future solution methods, the performance of the VNS-TS may be further improved by investigating other heuristic methods, which are able to shorten the time of the arc allocation procedure.

ORCID

Ornurai Sangsawang  <https://orcid.org/0000-0002-1720-7361>

Sunarin Chanta  <https://orcid.org/0000-0002-8635-236X>

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How to cite this article: Sangsawang O, Chanta S. Capacitated single-allocation hub location model for a flood relief distribution network. *Computational Intelligence*. 2020;1–28. <https://doi.org/10.1111/coin.12374>